

Task Context
You are an expert in software configuration analysis. Your task is to carefully read the provided software manual, analyze the dependency relationships between configuration options described in the documentation, and output the identified dependencies using a clearly defined BNF grammar format.
Reasoning Instructions (with symbolic knowledge)
When given a piece of documentation, follow these steps:
1. Identify the most relevant descriptions of configuration options. 2. Analyze how these configuration options influence each other based on descriptions such as default values, valid conditions, usage constraints, or fallback behaviors. 3. Express the dependency using the given BNF grammar rules.
If there is no dependency between the two configuration options, output "None"; otherwise, provide your output in the following format:
Formula: $\langle \text{Dependency} \rangle$ Explanation: "Reasoning behind analysis."
The BNF grammar you should follow to describe the configuration dependencies is as follows: <code>see Table 1</code> .
Example 1:
Doc snippet: <code>Descriptions for options O_x and O_y, $\mathcal{D}_x$ and $\mathcal{D}_y$</code> Output: <code>BNF of dependency between options O_x and O_y</code>
The Actual Task (with symbolic knowledge)
Now, please analyze options O_α and O_β in the provided descriptions and output the configuration dependency using the format demonstrated above.
Configuration option name O_α: <code>O_α</code> Configuration option name O_β: <code>O_β</code> Description for O_α: <code>$\mathcal{D}_\alpha$</code> Description for O_β: <code>$\mathcal{D}_\beta$</code>

Figure 1: Prompt template for LLM-symbolic dependency formalization via manuals.